

materials?

1. recall the key features of a life cycle assessment (LCA) including:
 - the main requirements for energy input;
 - the environmental impact and sustainability of making the material from natural resources;
 - the environmental impact of making the product from the material;
 - the environmental impact of using the product;
 - the environmental impact of disposing of the product by incineration, landfill or recycling;
2. **understand how the outcomes of a Life Cycle Assessment (LCA) for a particular material will depend on which product is made from the material;**
3. when given appropriate information relating to a Life Cycle Assessment (LCA), compare and evaluate:
 - the use of different materials for the same job;
 - the use of the same material for different jobs.
4. in the context of a Life Cycle Assessment:
 - can distinguish questions which could be addressed using a scientific approach, from questions which could not;
 - can identify the groups affected and the main benefits and costs of a course of action for each group;
 - can explain the idea of sustainable development, and apply it to specific situations;
 - shows awareness that scientific research and applications are subject to official regulations and laws;
 - **can distinguish between what can be done (technical feasibility) and what should be done (values);**
 - **can explain why different courses of action may be taken in different social and economic contexts.**

This module follows the commercial ‘food chain’ from farm to plate. Intensive and organic farmers use different methods to maintain soil fertility by recycling chemicals. Farmers also use a range of techniques to combat loss of crop yields by competition from weeds, or attack by pests and diseases.

There may be harmful or toxic chemicals in the food we eat. Some occur naturally. Some are deliberate additives. The added chemicals may be to preserve foods or to improve their colour, texture and flavour. These chemicals need not be harmful in small amounts. Their effects depend on how much we eat. To determine the safe levels of chemicals in food it is necessary to carry out a risk assessment. Regulators ensure that food does not contain any chemicals known to be unsafe.

Our bodies digest the food we eat. Digestion breaks down natural polymers such as starch and protein. The smaller molecules produced can be absorbed into the bloodstream. Some processed foods contain a high level of sugar which enters the bloodstream quickly. Insulin is available to treat people with diabetes who cannot control blood sugar levels normally.

Issues for citizens	Questions that science may help to answer
Is organic food better for us?	What is the difference between intensive and organic farming?
What are food additives, and why are they used?	Why are chemicals deliberately added to food?
Are food additives safe to eat?	How can we make sure that our food does not contain chemicals that may be harmful to health?
Why can it be harmful to eat too much sugary food?	Why does what we eat affect our health?
Science Explanations	Ideas about Science
SE 5 b,c The chemical cycles of life	IaS 5.1–5.5 Risk
SE 7 a,d Maintenance of life	IaS 6.1–6.3, 6.7 Making decisions about science and technology

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science. For example, disseminating scientific findings to the public in forms which allow individuals to make decisions about the issues related to food safety.

Use of ICT in teaching and learning can include:

- internet to research particular food additives;
- modelling software to display small and large biological molecules;
- animation to illustrate key stages in the nitrogen cycle.